

RAPHAEL HUANG – CURRICULUM VITAE

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RESEARCH INTERESTS

FPGA/ASIC design and verification; formal verification of hardware systems; hardware security, side-channel analysis; VLSI, place-and-route algorithms; heterogeneous computing for ML.

EDUCATION

University of California, Santa Cruz

September 2025 - Present

Master of Science in Computer Science and Engineering

Expected Graduation: December 2026

University of California, Santa Cruz

September 2021 - June 2025

Bachelor of Science, *magna cum laude* in Computer Engineering

GPA: 3.93

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant, UC Santa Cruz, Baskin Engineering

September 2025 - Present

- Coordinated logistical and technical aspects of a ~150 student Introduction to Logic Design course.
- Oversaw the creation of homework and exams, ensuring consistent evaluation of student learning.
- Lectured and held lab sections, debugging large structural Verilog designs in Xilinx Vivado one-on-one with students.

Graduate Research Aide, Argonne National Laboratory

June 2026 - September 2026

- Accelerated verification of Argonne's Chisel-based hardware accelerators by developing Mallet [F1], an open-source formal verification (FV) tool that derives properties from a design's AXI memory-map annotations and issues them to multiple backend model-checkers.
- Developed deep familiarity with the open-source FV stack, including Chisel/Scala, CIRCT/firtool compiler internals, temporal logic (SVA/LTL), and IC3/BMC engines (rIC3, pono, btormc), diagnosing a silent failure in CIRCT's BTOR2 emission path [P1].
- Extended openhwfp-eval, an automated hardware floating-point evaluation dataset, with an HDL-agnostic flow that validates and runs PPA benchmarks for 71 designs from across three open-source Chisel FP libraries using robust SW reference vector generation.

Connected and Automated Vehicle R&D Intern, Argonne National Laboratory

June 2025 - August 2025

- Transitioned LARS (Longitudinal Automated Replay System), a real-time vehicle energy analytics tool, from Simulink to a modular ROS 2 stack, enabling rapid cross-vehicle deployment [R2].
- Validated the system through simulation and on-road testing on a Cadillac LYRIQ, implementing an Iterative Learning Control algorithm that reduced root-mean-square tracking error by ~65%.

Instructional Design Research Assistant, UC Santa Cruz, Hardware Systems Collective

Jan - July 2025

- Integrated open-source EDA tools such as Yosys and Verilator into the curriculum for CSE 100, Introduction to Logic Design, to enable remote completion of coursework, enabling enrollment to scale to over 140 students per quarter (previously ~80).

Lead Instructor, iDTech Camps, Stanford University

July 2024 - August 2024

- Led 11 instructors teaching AI/ML in Python and VEX Robotics to 100+ students while adapting curriculum for a pilot program for international learners from China and Latin America.

RESEARCH AND PROJECTS

Characterizing FPGA Power Side-Channel Recovery Sensors January 2026 - Present

- Building a framework to quantify the granularity and dynamic range of different delay-line implementations for FPGA **power side-channel recovery** attacks.
- Running trace collection on **Zynq Ultrascale** architecture and applying statistical analysis techniques (e.g. t-tests, normality tests) to create taxonomy of various sensor designs.

Runtime Reconfigurable LUT-based Logic Gate Networks on FPGAs January 2026 - May 2026

- Developed a nanosecond-scale **neural network** inference architecture by mapping differentiable lookup-table networks to reconfigurable FPGA overlays [R3].
- Utilized **PYNQ** architecture to bypass hardware synthesis and bitstream-loading bottlenecks.

Formal Methods for VLSI Timing Optimization September 2025 - January 2026

- Developed a formal methods approach to correct **Static Timing Analysis (STA)** violations by integrating a Z3 OMT solver with the **OpenROAD** open-source ASIC design flow.
- Created **Python** and **Tcl** scripts to parse timing reports, generate logical constraints for optimal cell resizing, and programmatically modify netlists [R1].

Vivaldi: A SystemVerilog FPGA Audio Synthesizer January - March 2025

- Wrote large-scale RTL in **SystemVerilog** for a Artix-7 FPGA to implement a multi-oscillator audio synthesizer; automated the verification and synthesis flow and validated the design with a comprehensive **DPI-C** testbench.

HONORS, AWARDS, AND FELLOWSHIPS

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| [A4] | 2026 | TLC Graduate Pedagogy Fellowship: Pedagogical Leadership Certificate, Teaching & Learning Center, University of California, Santa Cruz |
| [A3] | 2025 | Impact Argonne Award: Energy Secretary Visit, Argonne National Lab |
| [A2] | 2021-2025 | Dean's Honors, University of California, Santa Cruz |
| [A1] | 2021 | UC Regents Scholar, Regents of the University of California |

SCHOLARLY AND CREATIVE WORK

POSTERS (PEER REVIEWED)

- [F1] Huang, Raphael and Yoshii, Kazutomo, *Mallet: An Open-source Formal Verification Flow for Agile Domain-Specific Accelerator Development*, ACM/IEEE International Conference for High Performance Computing, Networking, Storage and Analysis (SC), Argonne National Laboratory, 2026.

TECHNICAL REPORTS

- [R1] C. Baker, R. Huang, W. Ali, A. Muddamsetti, *Optimal Cell Resizing in OpenROAD-flow-scripts Using Optimization Modulo Theories*, Baskin Engineering, University of California, Santa Cruz, 2025.
- [R2] Goberville, Nick and Huang, Raphael, *Modernizing LARS: Open-source, Modular, and Programmatically Flexible Scenario Replay*, Argonne National Lab, 2025.
- [R3] R. Huang, C. Baker, M. Frank, *LiveLLNN: Reconfigurable, Heterogeneous Lookup Table-Based Logical Neural Networks*, 2026.

CONFERENCE AND WORKSHOP PRESENTATIONS

- [P1] *Open-source Hardware Verification in the Age of Agents*, ARCHIDE: 3rd Workshop on Architecture Design Methodologies, Ecosystems for HPC, and Scientific Edge Computing, 2026.